



Haichao Zhang

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[Google Scholar](#) | [GitHub](#) | [Personal Website](#)

Education

University of Liverpool , Ph.D. in Artificial Intelligence and Large Language Models	2024.06 – Present
Research interests include large language models (LLMs), foundation models, retrieval-augmented generation (RAG), generative reasoning, and model optimization.	
Xi'an Jiaotong-Liverpool University , M.Sc. in Computer Science	2022.09 – 2024.03
Graduated with Distinction; ranked in the top 1% of the program.	
East China Jiaotong University , B.Eng. in Computer Science and Technology	2014.09 – 2018.06
Average score: 87/100; ranked in the top 1% of the program.	

Work Experience

Alibaba (Beijing) Software Technology Co., Ltd. , Development Engineer	2018.11 – 2022.08
- Developed data platforms for Alibaba's advertising delivery and recommendation systems, covering core components such as the audience engine, inspection and attribution, and tag computation. - Contributed to large-scale data platforms, complex business systems, and core engine platforms, with hands-on experience in high-concurrency system development and optimization for complex scenarios.	
Dingfu Data Technology (Beijing) Co., Ltd. , Development Engineer	2018.07 – 2018.11
Contributed to computer-vision algorithm development and image-processing projects, with a focus on image-to-table conversion and watermark removal.	

Academic Research

Customized Retrieval-Augmented Generation with Large Language Models	2025.01 – 2025.05
- First-author paper, <i>Customized Retrieval-Augmented Generation with LLM for Debiasing Recommendation Unlearning</i>, accepted by ICDM 2025 (CCF-B). - Proposed CRAGRU, a framework that integrates large language models with retrieval-augmented generation (RAG) to enable fine-grained knowledge retrieval, context construction, and generative reasoning; validated its effectiveness on multiple benchmark datasets. Project page: https://zhang-haichao.github.io/CRAGRU-Page - Research involved technologies including LLMs, RAG, retrieval augmentation, context modeling, and generative reasoning.	
Parameter-Efficient Model Adaptation via Prompt Tuning and Knowledge Distillation	2025.04 – 2025.08
- First-author paper, <i>Distilled Prompt Tuning for Efficient GNN-based Recommendation Unlearning</i>, currently under major revision at TOIS. - Combined dual-teacher knowledge distillation with parameter-efficient prompt tuning to enable lightweight adaptation of specific model behaviors without retraining the backbone model. - Research involved technologies including PEFT, prompt learning, knowledge distillation, and parameter-efficient adaptation.	
Controllable Generation via Inference-Time Control	2025.10 – 2026.01
- First-author paper, <i>Controllable Generative Recommendation via Guided Token Refinement</i>, accepted by ICDM 2026 (CCF-B). - Proposed ReGen, a controllable generation framework that reformulates the generation process as parallel iterative refinement over Semantic IDs, enabling real-time control during inference. - Research involved technologies including Transformers, generative modeling, inference-time control, parallel	

decoding, and constrained generation, achieving a balance between accuracy and diversity.

Sequential Modeling via Transformers and Causal Representation Disentanglement

2025.03 – 2025.09

- First-author paper, *Personalized Conformity Disentanglement for Debaised Recommendations*.
- Proposed PCDR, a framework that disentangles interest and conformity representations from users' dynamic behavioral sequences to improve individual-level modeling; **under review at JMLC**.
- Research involved technologies including Transformers, sequential modeling, attention mechanisms, causal inference, and contrastive learning; validated the method's effectiveness on multiple datasets.

Explain-then-Forget: Causal Explanation-based Unlearning for Efficient and Precise Recommendation

Recommendation Unlearning under Asymmetric Access Constraints

2026.01 – 2026.03

- First-author paper, *Explain, Then Forget: Recommendation Unlearning under Asymmetric Access Constraint*.
- Proposed CEU, a framework that achieves recommendation unlearning through causal contribution identification and localized ranking intervention without accessing or modifying model parameters; **accepted by ICDM 2026 (CCF-B)**.
- Integrated causal inference, counterfactual explanations, and recommender systems; validated the method's unlearning effectiveness, recommendation performance, and computational efficiency across three public datasets and multiple recommendation models.

Dual-Rate User Semantic Memory for LLM-Enhanced Sequential Recommendation

2026.03 – 2026.07

- First-author paper, *Dual-Rate User Semantic Memory for LLM-Enhanced Sequential Recommendation*.
- Proposed DRUMRec, a dual-rate semantic memory framework for efficient LLM-enhanced sequential recommendation that preserves rich user semantics while enabling lightweight online serving; **under review at AAAI 2026**.
- Research involved Transformer adaptation and training, cross-attention, joint optimization of large and lightweight models, and memory caching; validated the method's effectiveness on multiple datasets.

Publications and Research Outputs

- **Explain-then-Forget: Causal Explanation-based Unlearning for Efficient and Precise Recommendation**, ICDM 2026.
- **Controllable Generative Recommendation via Guided Token Refinement**, ICDM 2026.
- **Explaining Graph Recommendations via Counterfactual Support Sets**, ICONIP 2026.
- **Topology-Aware Inverse Preference Learning for Robust Hybrid Voting Systems**, ICONIP 2026.
- **Customized Retrieval-Augmented Generation with LLM for Debiasing Recommendation Unlearning**, ICDM 2025.
- **Machine Vision-Enabled Octahedral Network Reconstruction and Structural Analysis of Perovskite Quantum Dots**, ACS Nano.
- **Clustering-based Incremental Learning for Imbalanced Data Classification**, Knowledge-Based Systems.
- **Uncertainty-Aware Semantic Decoding for LLM-Based Sequential Recommendation**, APWeb 2025.
- **Counterfactual Contrastive Learning for Fine Grained Image Classification**, ICANN 2024.
- **Two-branch Network with Feature Fusion for Time Since Deposition Estimation of Bloodstains**, CSCWD 2024.

Intellectual Property

- **Invention Patent:** A Debiasing Method and Recommendation System Based on Personalized Causal Decomposition;
Publication No.: CN117312675A; **Publication Date:** December 29, 2023.
- **Software Copyright:** Unbiased Recommendation Algorithm System Based on Personalized Conformity Disentanglement V1.0;
Registration No.: 2025SR0516629; **Registration Date:** March 25, 2025.
- **Software Copyright:** Large-Scale Road-Network Taxi Dynamic Dispatch and Organization Platform for Mobile Internet Ridesharing; [YiBu V1.0]
Registration No.: 2018SR047738; **Registration Date:** January 22, 2018.
- **Software Copyright:** Android-Based Railway Equipment Inspection System; [Railway Inspection Equipment APP V1.0]

Registration No.: 2018SR717297; **Registration Date:** September 6, 2018.

- **Software Copyright:** Web-Based Railway Equipment Inspection System; [Railway Inspection Equipment System V1.0]

Registration No.: 2018SR643652; **Registration Date:** August 13, 2018.

Project Experience

Audience Engine Management Platform 2020.04 – 2022.09

- Built unified management, workflow visualization, and anomaly-localization capabilities for the core audience engine of Alibaba’s advertising data platform.
- Served as project lead, completing platform design, core feature development, and production delivery while coordinating interface design and platform iterations with engine owners.
- Established the system as the central management platform, significantly improving troubleshooting and operational efficiency.

Million-Scale Audience Inspection and Root-Cause Analysis Platform 2020.04 – 2022.09

- Built a closed-loop system for million-scale audience inspection, anomaly detection, root-cause analysis, and issue remediation across Alibaba’s core advertising-delivery pipeline.
- Served as project lead, designing the system architecture and core framework while coordinating multiple module owners to deliver the end-to-end solution.
- Supported dozens of business lines after launch and reduced customer complaint rates by more than 90%.

Real-Time Tag Computation and Audience Construction Engine 2019.07 – 2020.03

- Designed the real-time tag-computation and audience-construction engine for advertising-delivery scenarios and developed its core modules.
- Optimized the tag-production pipeline using graph computation, covering the computation kernel, index design, DSL definitions, and other critical modules.
- Reduced tag iteration time from 7 days to 0.5 days and saved approximately 70% of computing resources.

Skills

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- **Programming Languages:** Python (4 years), Java (6 years).
 - **Deep Learning Frameworks:** Proficient in PyTorch; capable of independently completing model design, training, evaluation, and optimization.
 - **Large Language Model Technologies:** Familiar with LLMs, RAG, Transformers, prompt tuning, knowledge distillation, model inference, and parameter editing.
 - **Systems and Data Engineering:** Experience developing big-data and application systems with Flink, MapReduce, HBase, Spring Cloud, and MySQL.
 - **Research:** End-to-end research experience spanning problem formulation, algorithm design, experimental validation, and academic writing.
 - **Languages and Certifications:** LanguageCert IESOL C1 Pass; Intermediate Professional Title / Software Designer.

Honors and Awards

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- First Prize, East China Jiaotong University ACM Programming Competition (2015.11).
 - Provincial Second Prize, Jiangxi Computer Works Competition and “Internet+” Innovation and Entrepreneurship Competition (2016.06).
 - First Prize, National Transportation Universities Outstanding Student Paper Competition (2016.09).
 - National Third Prize, China University Transportation College Maker Competition and “Internet Transportation” Innovation and Entrepreneurship Competition (2016.11).

Professional Summary

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- Combines industrial-scale system development with academic research, including experience building large-scale data platforms and high-concurrency systems at Alibaba as well as sustained research output in large language models.
 - Research focuses on LLMs, RAG, controllable generation, knowledge editing, and parameter-efficient adaptation, with end-to-end capability from problem formulation and algorithm design to experimental validation.
 - Strong programming foundation with the ability to independently drive complex projects, collaborate across teams, and deliver reliable results in demanding environments.